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SOFTWARE REQUIREMENTS SPECIFICATION

File Transfer System Using Elliptic Curve Cryptography

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the requirements for the File Transfer System Using Elliptic Curve Cryptography (ECC). The system is intended to provide secure file sharing between authenticated users by combining public-key cryptography with symmetric encryption.

1.2 Scope

The system provides a web-based mechanism for securely transferring documents, images, PDFs, text files, ZIP archives, and other digital files between authenticated users. ECC is used for secure key establishment or protection of a session key, while AES is used to encrypt the actual file efficiently. The initial implementation can be deployed on a local network or server.

The project can later be extended with cloud storage, multiple recipients, digital signatures, two-factor authentication, expiring download links, and mobile support.

1.3 Intended Users

The system is designed for authenticated users who need to send and receive files securely. The project is also intended to demonstrate practical application of cryptography, secure networking, Python web development, and database technologies.

1.4 Definitions and Acronyms

Term	Meaning
ECC	Elliptic Curve Cryptography
AES	Advanced Encryption Standard
SHA-256	Secure Hash Algorithm with 256-bit output
SRS	Software Requirements Specification
Session Key	Temporary symmetric key used to encrypt file data

1.5 References

- ❑ The project overview document supplied for the File Transfer System Using Elliptic Curve Cryptography.
- ❑ The reference SRS format supplied for Secure Data Transfer Over Internet Using Image Steganography.

2. Overall Description

2.1 Product Perspective

The proposed product is a web-based secure file-transfer application consisting of a user interface, backend processing, cryptographic components, database/storage, and network communication. Registered users authenticate themselves before using protected transfer functions.

2.2 Main Product Functions

- ❑ User registration and authentication.
- ❑ ECC public/private key generation.
- ❑ Secure key exchange or session-key protection.
- ❑ AES-based file encryption.
- ❑ Encrypted file transfer.
- ❑ File decryption by the authorized recipient.
- ❑ Integrity verification.
- ❑ Secure file download.
- ❑ Transfer history.

2. Overall Description (continued)

2.3 User Characteristics

Users are expected to have basic computer or smartphone skills, including logging in, selecting files, uploading or downloading data, and using a web application. No specialist cryptography knowledge is required for normal use.

2.4 Operating Environment

- ☐ Web application accessible through a modern web browser.
- ☐ Server capable of running the selected Python backend framework and cryptographic libraries.
- ☐ Database such as SQLite or MySQL for application data.
- ☐ Network connection for transferring encrypted files between users.

2.5 Assumptions and Dependencies

- ☐ Users have access to the application and a functioning network connection.
- ☐ Registered users have valid credentials.
- ☐ The receiver has the appropriate ECC private key required for key recovery.
- ☐ Cryptographic keys are protected from unauthorized access.
- ☐ The server and database remain available during a transfer.

3. System Overview

3.1 System Workflow

The system follows a sender-to-receiver workflow. Authentication is performed before protected operations. A random AES session key is generated for file encryption, while ECC is used to establish a shared secret or securely protect the AES session key. The encrypted file and required cryptographic information are transferred to the recipient. The receiver recovers the session key, decrypts the file, verifies integrity, and accesses the original file.

Stage	System Operation
1	Sender authenticates and selects a file.
2	System generates an AES session key.
3	File is encrypted using AES.
4	ECC key pair/key agreement protects or establishes the AES session key.
5	Encrypted file and protected key information are transferred.
6	Receiver authenticates and recovers the AES session key using ECC.
7	Receiver decrypts the file using AES.
8	System verifies file integrity and permits authorized download.

FILE TRANSFER SYSTEM USING ECC

SECURE FILE TRANSFER WORKFLOW

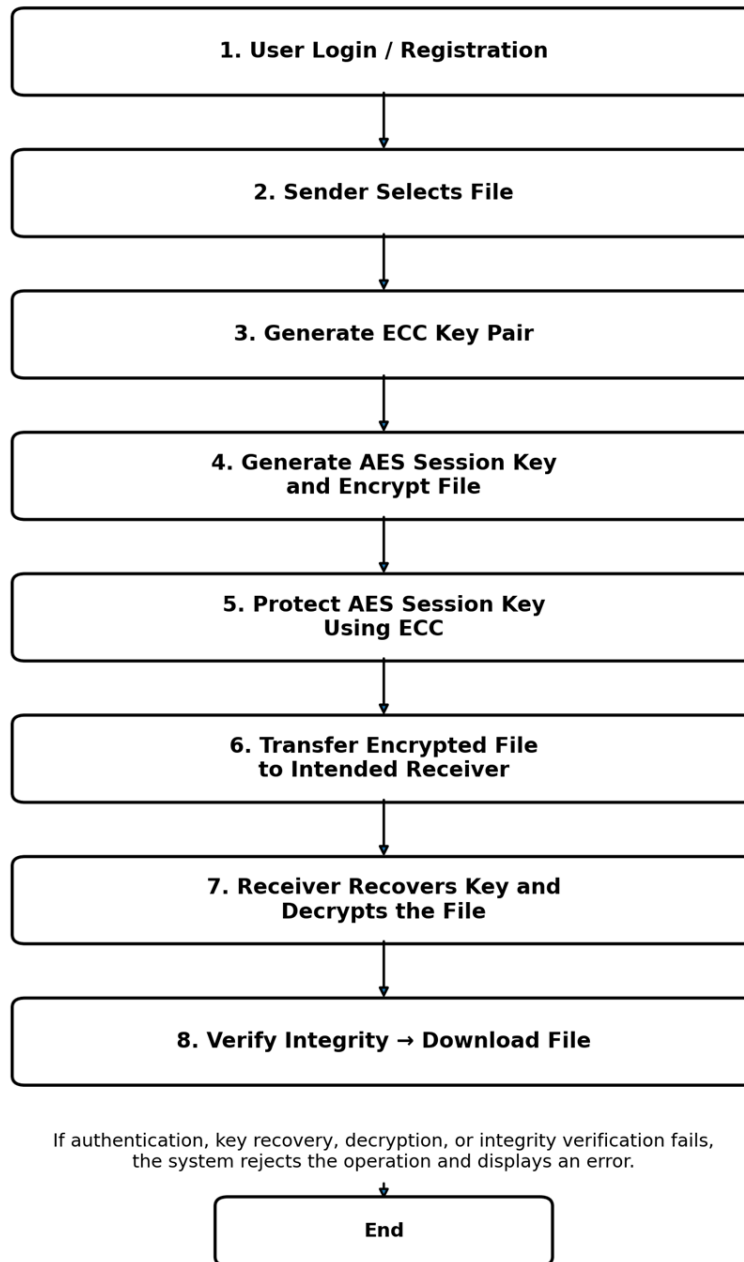


Figure: Simplified secure file-transfer workflow

3.2 System Context

Sender	Secure File Transfer System	Receiver
Authenticate, select, encrypt	Key management, transfer, storage, validation	Authenticate, recover key, decrypt, download
Send encrypted data	Network communication	Receive encrypted data

3. System Overview (continued)

3.3 System Architecture / Flow Chart

The original project document contains the system flow chart showing authentication, ECC key generation/recovery, AES session-key generation, encryption/decryption, integrity verification, and authorized download.

3.4 Actors and Events

Actor	Responsibilities
Sender	Registers/logs in, selects a file, initiates secure transfer, and sends the encrypted data to the intended recipient.
Receiver	Logs in, receives the encrypted file, recovers the session key, decrypts the file, verifies it, and downloads it.
System/Server	Authenticates users, generates/manages cryptographic keys, encrypts/decrypts, transfers/stores data, and performs integrity verification.

Key Events

#	Actor	Event	System Response
1	Sender	Logs in and selects file	Authenticates user and prepares transfer
2	Sender	Initiates transfer	Generates AES session key and encrypts file
3	System	Protects session key	Uses ECC key operations for secure key establishment/protection
4	Receiver	Receives encrypted transfer	Validates recipient access
5	Receiver	Requests file recovery	Recovers AES session key and decrypts file
6	System	Completes verification	Allows download if integrity is valid; otherwise rejects

4. Functional Requirements

4.1 Authentication Requirements

ID	Requirement
FR-01	The system shall allow users to register an account.
FR-02	The system shall authenticate registered users before protected functions are accessed.
FR-03	The system shall provide access control for protected file-transfer operations.

4.2 Sender Requirements

ID	Requirement
FR-04	The system shall allow the sender to select a file for transfer.
FR-05	The system shall generate an ECC public/private key pair for users as required by the key-management process.
FR-06	The system shall generate a random AES session key for file encryption.
FR-07	The system shall encrypt the selected file using AES before transmission.
FR-08	The system shall use ECC for secure key establishment or protection of the AES session key.
FR-09	The system shall transfer the encrypted file and required protected key information to the intended recipient.

4.3 Receiver Requirements

ID	Requirement
FR-10	The system shall allow the intended receiver to access the transferred encrypted file.
FR-11	The system shall use the receiver's appropriate ECC private-key operation for session-key recovery.
FR-12	The system shall decrypt the transferred file using the recovered AES session key.
FR-13	The system shall verify the integrity of the received file.
FR-14	The system shall allow the receiver to download the original file after successful verification.
FR-15	The system shall deny access to unauthorized recipients.

4.4 Transfer Management Requirements

- ⑦ The system shall maintain relevant information about previous transfers as transfer history.
- ⑦ The system shall protect cryptographic key information during transfer and storage.
- ⑦ The system shall report failed operations without presenting an invalid transfer as successful.

5. Use-Case Overview

5.1 Use-Case Diagram

Actor	Use Case	Description
Sender	Register / Login	Create an account or authenticate before protected operations.
Sender	Select File	Choose the file to be securely transferred.
Sender	Create Secure Transfer	Generate AES key, encrypt file, protect session key using ECC, and transfer data.
Receiver	Login	Authenticate as a registered user.
Receiver	Receive Transfer	Obtain the encrypted file and protected key information.
Receiver	Recover and Decrypt	Recover the AES key using ECC and decrypt the file.
Receiver	Verify and Download	Verify integrity and download the original file.

5.2 Use-Case List

Use Case	Primary Actor	Description
Create Secure Transfer	Sender	Encrypts a selected file using AES and securely protects/establishes its session key using ECC.
Receive Transfer	Receiver	Receives the encrypted file and associated protected key information.
Recover File	Receiver	Recovers the AES session key, decrypts the file, verifies integrity, and downloads the original.

5.3 Detailed Use-Case: Create Secure Transfer

Field	Details
Use-Case ID	UC-01
Use-Case Name	Create Secure Transfer
Primary Actor	Sender
Description	The sender selects a file and creates a secure transfer using AES encryption and ECC-based key management.
Precondition	The sender is registered/authenticated and has a file to transfer.
Main Flow	1. Sender logs in. 2. Selects file. 3. System generates AES session key. 4. File is encrypted using AES. 5. ECC protects/establishes the session key. 6. Encrypted data is transferred to the intended receiver.
Alternative Flow	If authentication, encryption, key management, or transfer fails, the system reports an error and does not present the operation as successful.
Postcondition	An encrypted file transfer is created for the intended recipient.

5. Use-Case Overview (continued)

5.4 Detailed Use-Case: Receive Transfer

Field	Details
Use-Case ID	UC-02
Use-Case Name	Receive Transfer
Primary Actor	Receiver
Description	The receiver authenticates and obtains the encrypted file and the required protected key information.
Precondition	The receiver is registered/authenticated and is an intended recipient.
Main Flow	1. Receiver logs in. 2. System validates access. 3. Receiver receives the encrypted transfer. 4. Protected key information is made available for recovery.
Alternative Flow	If the receiver is unauthorized or the transfer is unavailable, access is denied and an error is displayed.
Postcondition	The encrypted file is available for recovery and decryption.

5.5 Detailed Use-Case: Recover File

Field	Details
Use-Case ID	UC-03
Use-Case Name	Recover File
Primary Actor	Receiver
Description	The receiver recovers the AES session key using ECC, decrypts the file, verifies integrity, and downloads the original file.
Precondition	The receiver has authorized access to the encrypted transfer and the appropriate ECC private key.
Main Flow	1. Receiver requests recovery. 2. System performs ECC key recovery. 3. AES session key is recovered. 4. File is decrypted using AES. 5. Integrity is verified. 6. Original file is made available for download.
Alternative Flow	If key recovery, decryption, or integrity verification fails, the system rejects the operation and displays an error.
Postcondition	The receiver downloads the verified original file.

6. External Interface Requirements

6.1 User Interface

- ❑ The interface shall provide registration and login controls.
- ❑ The interface shall provide a file-selection control for the sender.
- ❑ The interface shall display transfer and processing status.
- ❑ The interface shall provide clear success and error notifications.
- ❑ The interface should be usable on common desktop and mobile-sized screens.

6.2 Input Interfaces

- ❑ User registration/login credentials.
- ❑ File selected by the sender.
- ❑ Recipient/transfer information as required by the implementation.
- ❑ Cryptographic key information managed by the application.

6. External Interface Requirements (continued)

6.3 Output Interfaces

- ☐ Encrypted file/transfer status.
- ☐ Success and error messages.
- ☐ Decrypted original file available for authorized download.
- ☐ Transfer history information.

6.4 Software Interfaces

The application may use a Python web backend, Flask web framework, HTML5, CSS3, JavaScript, a cryptographic library for ECC and AES operations, and SQLite or MySQL for database management. Git/GitHub may be used for version control.

7. Non-Functional Requirements

Category	Requirement
Security	Files shall be encrypted before transmission. ECC shall be used for secure key establishment or session-key protection.
Usability	The system shall provide a simple workflow and understandable instructions for authenticated users.
Performance	The system should perform encryption, transfer, decryption, and download within a reasonable time for supported file sizes.
Reliability	The system shall report failed operations without producing an apparently successful but unusable transfer.
Availability	The application shall be available when required, subject to server and network availability.
Compatibility	The web application should work with commonly used modern browsers.
Maintainability	The system should be organized into separate authentication, processing, cryptographic, storage, and interface components.

Technologies Used

Technology	Purpose
Python	Backend programming
Flask	Web application framework
HTML5	Web page structure
CSS3	User interface styling
JavaScript	Client-side functionality
ECC	Secure key establishment / key protection
AES	Efficient file encryption
SHA-256 / authenticated encryption	Integrity and authenticity support
SQLite / MySQL	Database management
Git / GitHub	Version control

8. Security Requirements

- ❑ The system shall encrypt files before transmission.
- ❑ The system shall use ECC for secure public-key operations and session-key management.
- ❑ Private keys shall be protected from unauthorized access.
- ❑ The system shall verify the integrity of transferred data.
- ❑ The system shall provide authentication and access control.
- ❑ The system shall deny unauthorized recipients access to protected files.
- ❑ Cryptographic information shall not be unnecessarily exposed through application messages or transfer metadata.
- ❑ Secure communication such as HTTPS should be used when deployed for actual Internet use.

9. Constraints and Assumptions

Type	Description
Constraint	Encryption and decryption introduce processing overhead.
Constraint	Private-key protection and key management must be implemented correctly.
Constraint	Network connectivity is required for remote transfer.
Constraint	Very large files may require additional storage and processing resources.
Constraint	Overall security depends on correct implementation and secure configuration.
Assumption	The receiver can recover the original file only when the required authorization and cryptographic key information are available.

10. Conclusion

This SRS defines the essential requirements for a web-based File Transfer System Using Elliptic Curve Cryptography. By combining ECC for secure key management with AES for efficient file encryption, the system provides a practical approach to confidentiality, secure key handling, integrity verification, authentication, and access control. The requirements in this document can serve as the basis for design, implementation, testing, and evaluation of the project.